

Lecture Week 1: ALGEBRAIC FUNCTIONS

(REVIEW of HIGH SCHOOL MATHEMATICS)

1.1. Definition of a Function.

A function f from set A to set B (or simply function) is a rule that assigns to each element (or number) x of the domain set A exactly one element called $f(x)$ of the codomain set B .

We can also say that

- function f is an assignment that associates to each element in the set A exactly one element of set B
- function f maps elements from the set A into elements in set B .
- $y = f(x)$ is the image of x under transformation f .
- function f expresses a relationship between two sets of numbers.

We'll be using notation " f from A to B " $f: A \rightarrow B$ or $y = f(x)$

x is called independent variable, y is called dependent variable

There are four ways to represent the function: tables, diagrams, formulas, and graphs.
We'll be using notation R to define the set of real numbers.

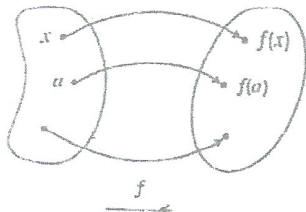
The domain of f is the largest subset of R , for which f is defined:

$$Dom(f) = \{x \in R \mid f(x) \text{ is defined}\}$$

The range of f is the set of all possible values $y = f(x)$:

$$Ran(f) = \{y \in R \mid y = f(x) \text{ for } x \in Dom(f)\}$$

Notations used in definitions are called "set notations"; " $\mid$ " means "such that", " $\in$ " means "contains in"



Machine diagram for a function f

Example:

- Consider function $f = \sqrt{x}$. It's defined for all positive numbers. All positive numbers form the set A , which is proper subset of R . In the set notation: $A \subset R$. So, the set A in this example is the domain of $f = \sqrt{x}$. $f(x)$ has domain $[0, \infty]$
- Consider function $f = x^2$. It's defined for all real numbers; so all real numbers form the set A . The output of this function is the set B that can contain only positive numbers. The set B is a proper subset of R ; $B \subset R$ where the set R is the codomain of f and the set B is the range of $f = x^2$. $f(x)$ has range $[0, \infty]$.
- Consider function $f = x^3$. This function is defined for all real numbers and its output is also the set of all real numbers. In this example codomain of f is also its range.

Notations used in example are called "interval notations".

Example: Express Domain and Range of $f(x) = \frac{1}{\sqrt{x^2-1}}$ in interval notation.

1.2. Equality between functions.

Two functions are equal iff:

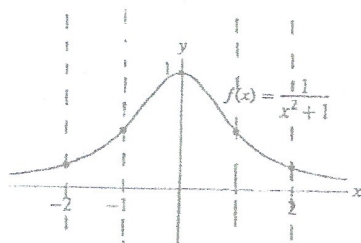
- a) $f(x) = g(x)$ for all x in the domain of f ;
- b) $f(x)$ and $g(x)$ defined on the same domain.

Example: Whether or not $f(x) = \frac{x^2-1}{x-1}$ is equal to $g(x) = x+1$?

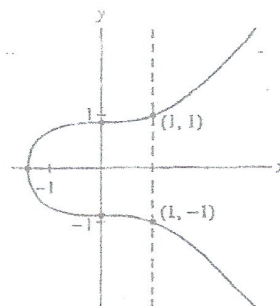
1.3. Vertical Line Test (explicit $f(x)$). $y = f(x)$

A curve in the XY plane is the graph of a function iff no vertical line intercepts the curve more than once.

Function.



Not a Function.



Graph of $4y^2 - x^3 = 3$. This graph fails the Vertical Line Test, so it is not the graph of a function.

1.4 Even and Odd Functions.

A function f is even if $f(-x) = f(x)$ for all x in the domain of $f(x)$.

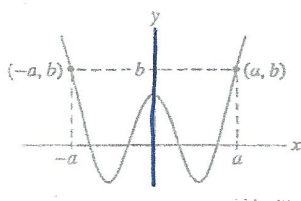
Graphs of even functions are symmetric with respect to y -axis.

Example: $f(x) = x^2 - 3$ is the even function. Check $f(-x) = (-x)^2 - 3 = x^2 - 3 = f(x)$

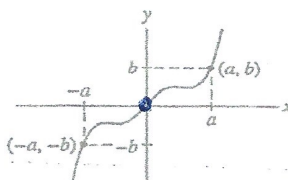
A function f is odd if $f(-x) = -f(x)$ for all x in the domain of $f(x)$.

Graphs of odd functions are symmetric with respect to the origin.

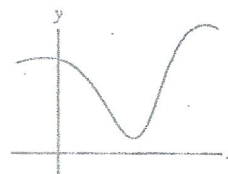
Example: $f(x) = x^3$ is the odd function. Check $f(-x) = (-x)^3 = -x^3 = -f(x)$



(A) Even function: $f(-x) = f(x)$
Graph is symmetric about the y -axis.



(B) Odd function: $f(-x) = -f(x)$
Graph is symmetric about the origin.



(C) Neither even nor odd

A function is neither even nor odd if $f(-x) \neq f(x)$ and $f(-x) \neq -f(x)$

Example: $f(x) = \frac{x}{x^3 + 2}$ is neither even nor odd function.

Check: $f(-x) = \frac{-x}{(-x)^3 + 2} = -\frac{x}{x^3 - 2} \neq f(x)$;

$$f(-x) = \frac{-x}{(-x)^3 + 2} = -\frac{x}{x^3 - 2} \neq -f(x)$$

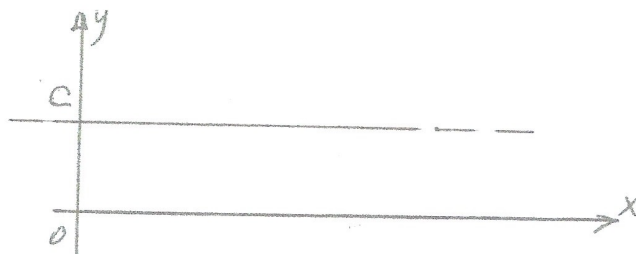
1.5 A Library of Algebraic Functions ($f(x) = y$).

1.5.1. Constant Functions

$$y = C$$

$$\text{Dom } f(x) := \{x \in \mathbb{R}\},$$

$$\text{Ran } f(x) := \{y \in \mathbb{R} | y = C\}$$



1.5.2. Power Functions

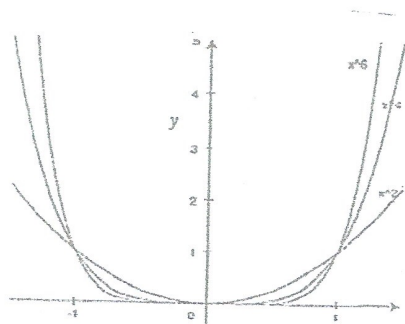
$$y = x^p$$

a) Even Power Functions ($p=2n, n=1,2,3,\dots$)

$$y = x^{2n}$$

$$\text{Dom } f(x) := \{x \in \mathbb{R}\},$$

$$\text{Ran } f(x) := \{y \in \mathbb{R} | y \geq 0\}$$

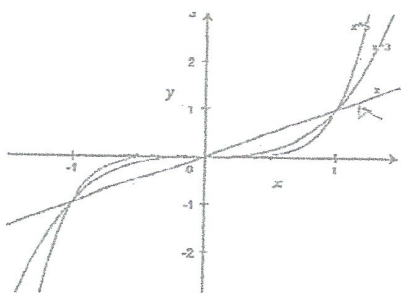


b) Odd Power Functions ($p=2n-1, n=1,2,3,\dots$)

$$y = x^{2n-1}$$

$$\text{Dom } f(x) := \{x \in \mathbb{R}\},$$

$$\text{Ran } f(x) := \{y \in \mathbb{R}\}$$

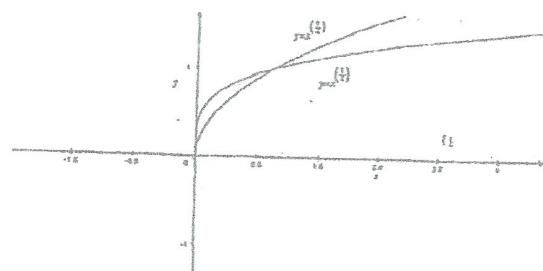


c) Even n -th Root Functions ($p = \frac{1}{2n}, n=1,2,3,\dots$)

$$y = x^{\frac{1}{2n}}$$

$$\text{Dom } f(x) := \{x \in \mathbb{R} | x \geq 0\},$$

$$\text{Ran } f(x) := \{y \in \mathbb{R} | y \geq 0\}$$

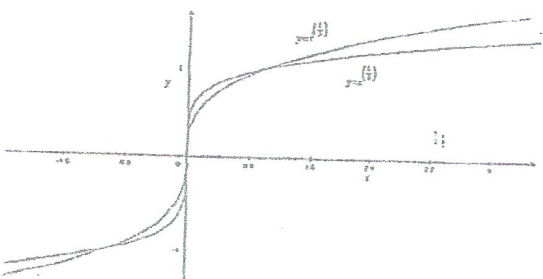


d) Odd n -th Root Functions ($p = \frac{1}{2n-1}, n=1,2,3,\dots$)

$$y = x^{\frac{1}{2n-1}}$$

$$\text{Dom } f(x) := \{x \in \mathbb{R}\},$$

$$\text{Ran } f(x) := \{y \in \mathbb{R}\}$$

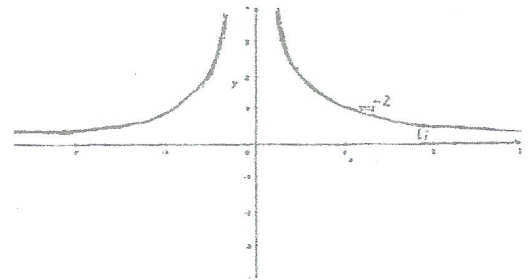


e) *Negative Even Power Functions* ($p = -2n, n=1,2,3\dots$)

$$y = x^{-2}$$

$$\text{Dom } f(x) := \{x \in \mathbb{R}\},$$

$$\text{Ran } f(x) := \{y \in \mathbb{R} | y > 0\}$$

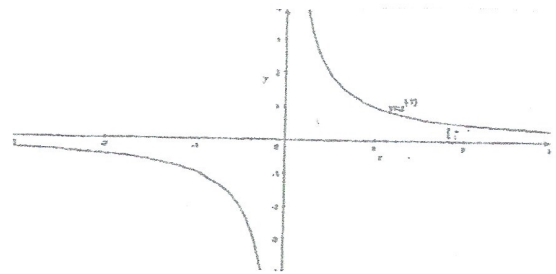


f) *Negative Odd Power Functions* ($p = -2n+1, n=1,2,3\dots$)

$$y = x^{-1}$$

$$\text{Dom } f(x) := \{x \in \mathbb{R} | x \neq 0\},$$

$$\text{Ran } f(x) := \{y \in \mathbb{R} | y \neq 0\}$$



1.5.3 Polynomial Functions.

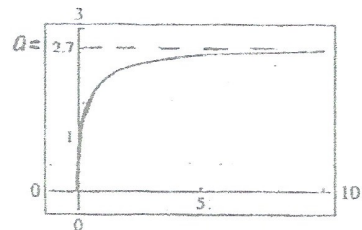
$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0 \quad \text{where } \{a_n, a_{n-1}, \dots, a_0\} \in \mathbb{R}$$

The degree of a polynomial function equals to the degree of leading term of corresponding polynomial: for $n=1$ we have linear function.

1.5.4. Rational Functions.

A rational function is any function of the type $f(x) = \frac{P_n(x)}{Q_m(x)}$ where $Q_m(x)$ is not the zero polynomial. The Domain of $f(x)$ is the set of all x such that $Q_m(x) \neq 0$.

Example. Monod Function $y(x) = a \frac{x}{b+x}$



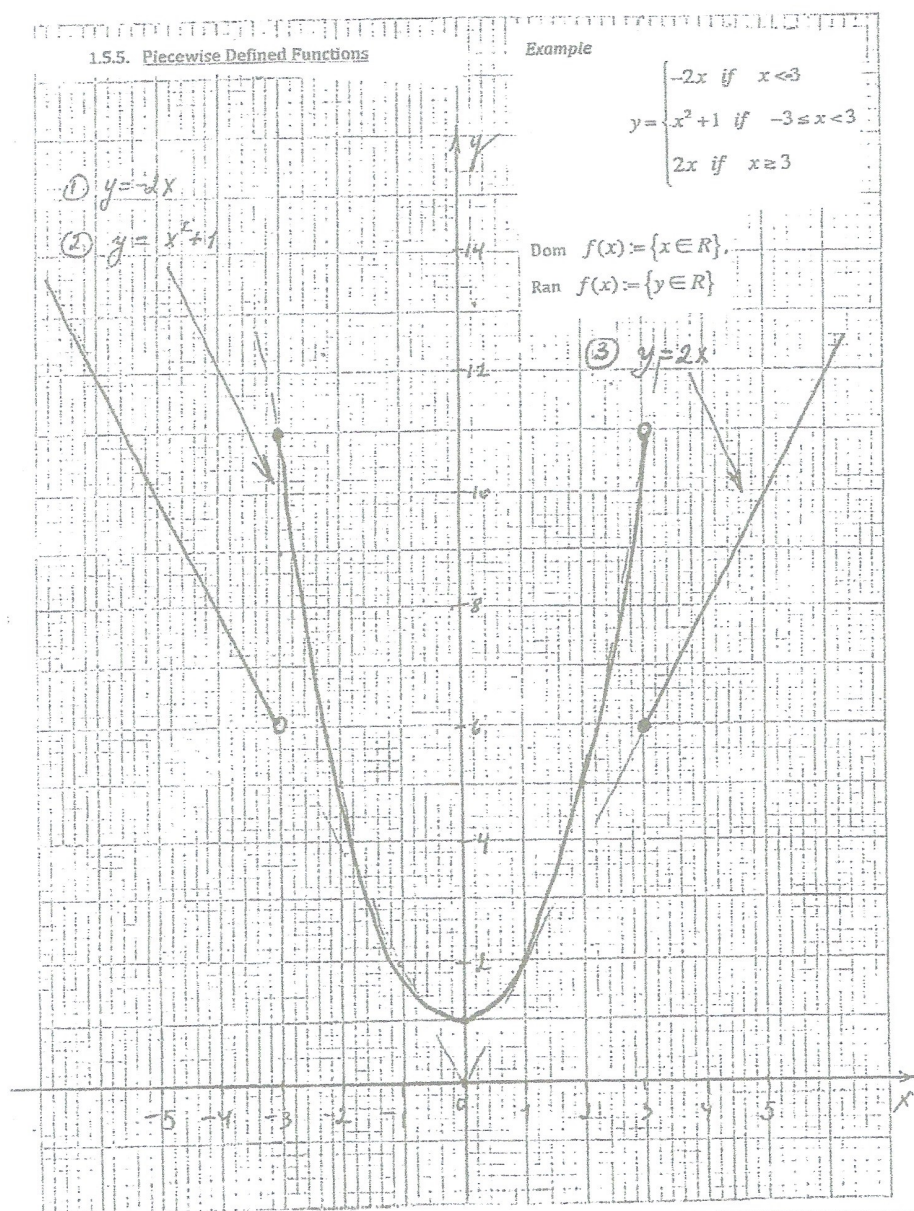
1.5.5. Piecewise Defined Functions

Example

$$y = \begin{cases} -2x & \text{if } x < -3 \\ x^2 + 1 & \text{if } -3 \leq x < 3 \\ 2x & \text{if } x \geq 3 \end{cases}$$

$$\text{Dom } f(x) := \{x \in \mathbb{R}\},$$

$$\text{Ran } f(x) := \{y \in \mathbb{R}\}$$



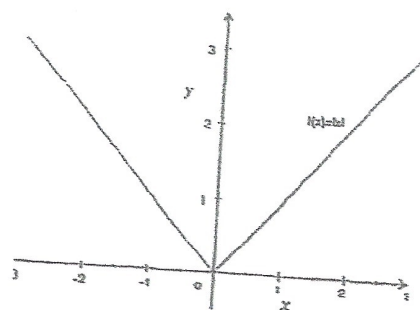
1.5.6. Absolute Value Functions.

Example

$$y = |x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

$$\text{Dom } f(x) := \{x \in \mathbb{R}\},$$

$$\text{Ran } f(x) := \{y \in \mathbb{R} \mid y \geq 0\}$$

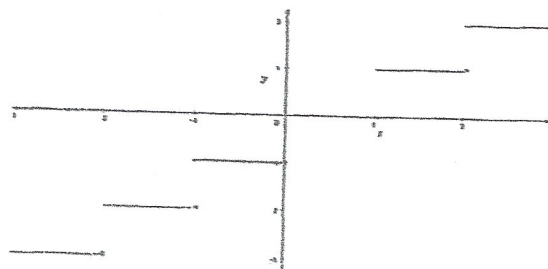


1.5.7 The Floor and Ceiling Functions

$$f(x) = \lfloor x \rfloor$$

$$\text{Dom } f(x) := \{x \in \mathbb{R}\},$$

$$\text{Ran } f(x) := \{y \in \mathbb{Z}\}$$



1.6 Operations with Functions.

Let f and g be functions with domains A and B respectively, then

$$(f \pm g)(x) = f(x) \pm g(x), \quad \text{Dom}(f \pm g) := A \cap B$$

$$(fg)(x) = f(x)g(x), \quad \text{Dom}(fg) := A \cap B$$

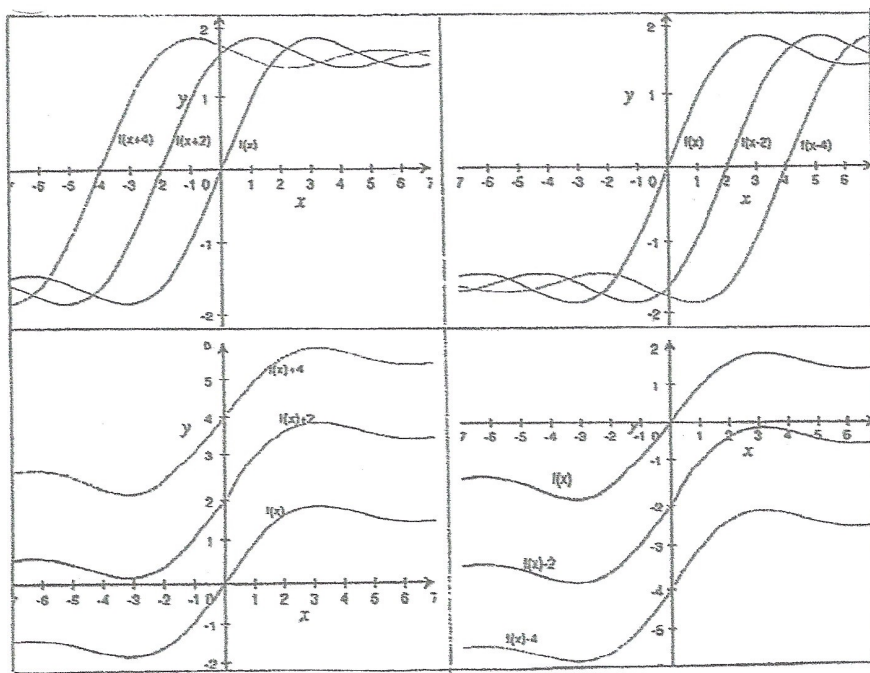
$$\frac{f}{g}(x) = \frac{f(x)}{g(x)}, \quad \text{Dom}\left(\frac{f}{g}\right) := \{x \in A \cap B \mid g(x) \neq 0\}$$

1.6.1 Transformations

a) Vertical and Horizontal Shifts.

Suppose $c > 0$, $c \in \mathbb{R}$ then to get the graph of

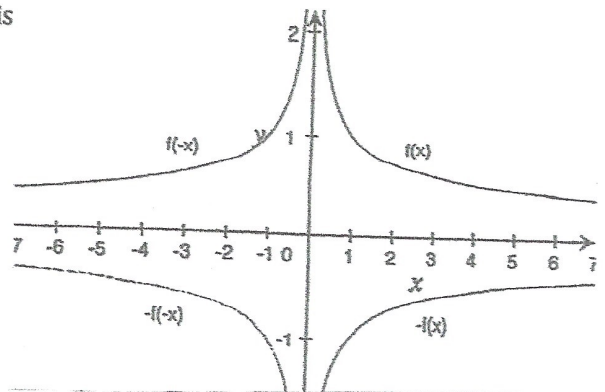
1. $y = f(x) + c$, shift the graph of $y = f(x)$ c units upward
2. $y = f(x) - c$, shift the graph of $y = f(x)$ c units downward
3. $y = f(x - c)$, shift the graph of $y = f(x)$ c units to the right
4. $y = f(x + c)$, shift the graph of $y = f(x)$ c units to the left



b) Reflecting

Suppose $c > 1$, $c \in \mathbb{R}$ then to get the graph of

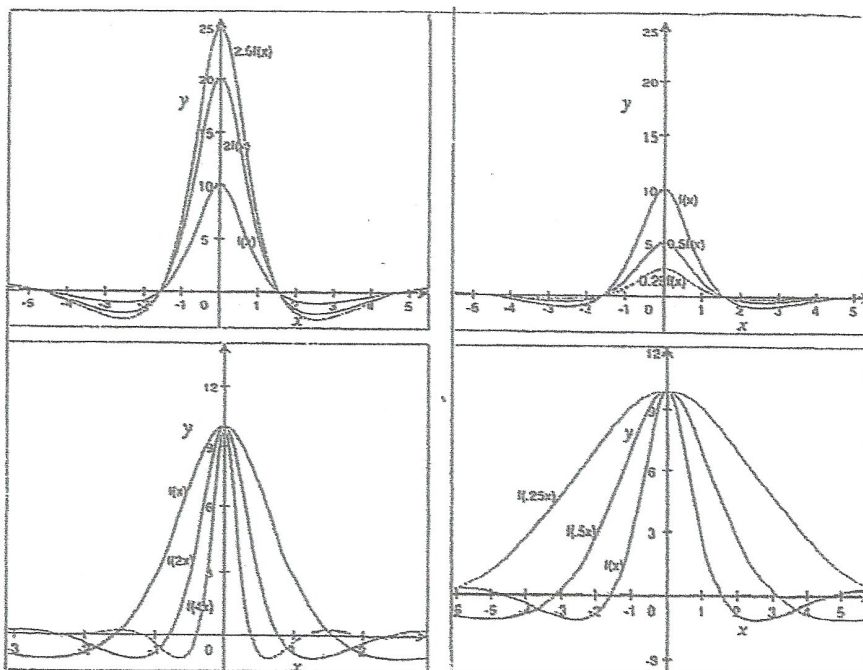
1. $y = -f(x)$, reflect the graph of $y = f(x)$ about the x -axis
2. $y = f(-x)$, reflect the graph of $y = f(x)$ about the y -axis



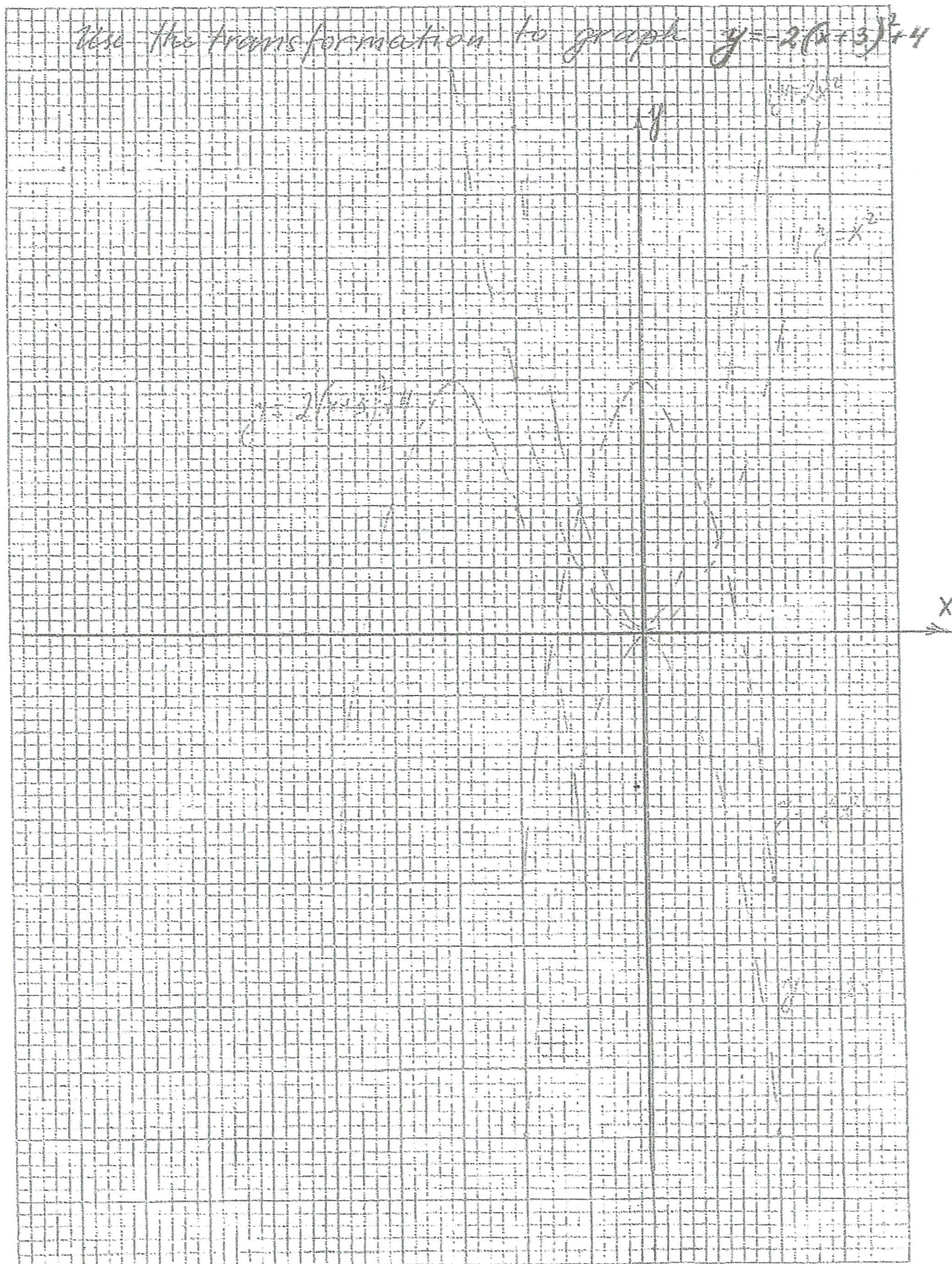
c) Vertical and Horizontal Stretching and Compression.

Suppose $c > 1$, $c \in \mathbb{R}$ then to get the graph of

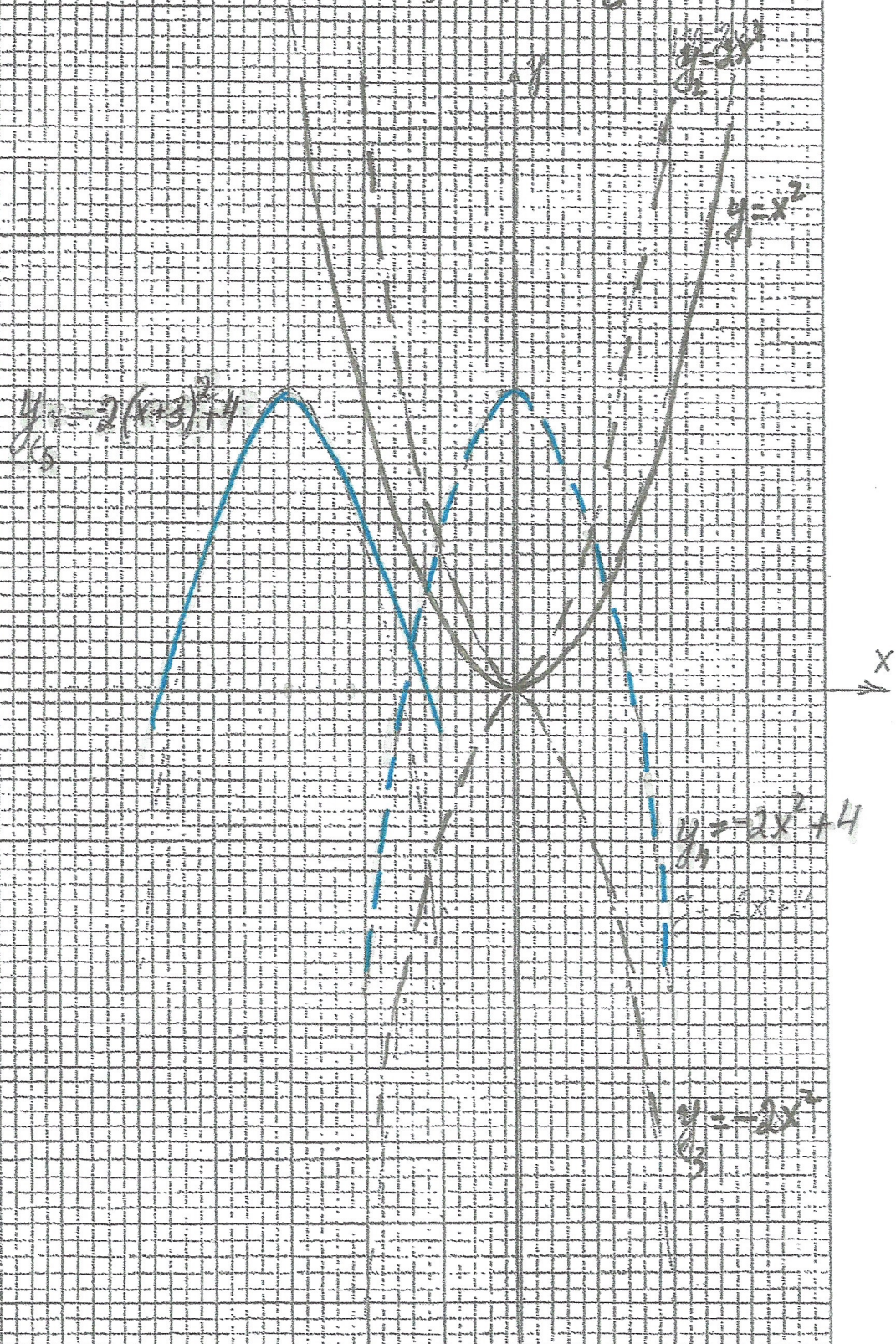
1. $y = cf(x)$, stretch the graph of $y = f(x)$ vertically by a factor c
2. $y = \frac{1}{c}f(x)$, compress the graph of $y = f(x)$ vertically by the factor c
3. $y = f(cx)$, compress the graph of $y = f(x)$ horizontally by the factor of c
4. $y = f\left(\frac{x}{c}\right)$, stretch the graph of $y = f(x)$ horizontally by a factor c



Use the transformation to graph $y = -2(x+3)^2 + 4$



Use the transformation to graph $y = 2(x+3)^2 + 4$



1.6.2. Composition of Functions

For two functions $f(x)$ and $g(x)$ composite function is defined by $f[g(x)] = f \circ g$.

Let D_f and D_g be domains of $f(x)$ and $g(x)$ respectively, then

the domain of $f[g(x)] = f \circ g$ is the set of all x in the domain of g such that $g(x)$ is the domain of f : $Dom(f \circ g) := \{x \in D_g \mid g(x) \in D_f\}$

Example:

Let $f(x) = x^2$ and $g(x) = \sqrt{x-3}$ Find $f \circ g$, $g \circ f$, $f \circ f$, $g \circ g$ and their domains.

Answer;

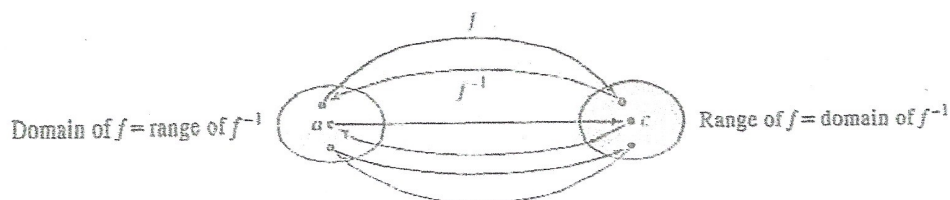
We also can apply the rule of composition to more than two functions. For example, the composition of three functions f, g and h is defined as $(f \circ g \circ h)(x) = f[g(h(x))]$

Example

Let $f(x) = \frac{1}{x}$, $g(x) = x^3$ and $h(x) = x^2 + 2$. Find $(f \circ g \circ h)(x) = f[g(h(x))]$

1.8 Inverse Functions.

Let f be a one-to one function with domain A and range B. Then its inverse function f^{-1} has domain B and range A and defined by $f^{-1}(y) = x \Leftrightarrow f(x) = y$



Example

Suppose for a given one-to one function f , $f(1) = 5$, $f(3) = 7$ and $f(8) = -10$.

Find $f^{-1}(5)$, $f^{-1}(7)$ and $f^{-1}(-10)$.

1.8.1 Property of Inverse Functions.

Let f be a one-to one function with domain A and range B. The following cancelation properties indicate that f^{-1} is the inverse of f and vice versa.

$$f^{-1}[f(x)] = x \text{ for all } x \text{ in } A$$

$$f[f^{-1}(x)] = x \text{ for all } x \text{ in } B$$

Example:

Show that $f(x) = x^3$ and $g(x) = \sqrt[3]{x}$ are inverses of each other.

Solution.

To show that these functions are inverses of each other we use the property on inverse functions. Note that domain and range of both functions is $\mathbb{R}$.

We have $g[f(x)] = g[x^3] = \sqrt[3]{x^3} = x$, so by the property of inverse functions f and g
 $f[g(x)] = f[x^{1/3}] = (\sqrt[3]{x})^3 = x$
 are inverses of each other.

1.8.2 Finding the Inverse Function of one-to one Rational Functions.

1. Write $y = f(x)$
2. Solve the equation for x in terms of y
3. Interchange x and y . The resulting function is $y = f^{-1}(x)$

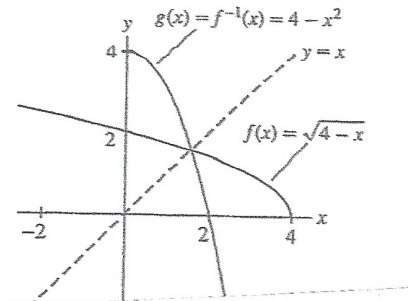
Example:

Find the inverse function of $f(x) = \sqrt{4-x}$ and sketch the graphs of $y = f(x)$ and $y = f^{-1}(x)$.

1. $y = \sqrt{4-x}$

2. $y^2 = 4-x$
 $x = 4-y^2$

3. $y = 4-x^2$



In order to sketch the graph, we observe that $Dom(f(x)) := \{x \in \mathbb{R} | x \leq 4\}$ and

$Ran(f(x)) := \{y \in \mathbb{R} | y \geq 0\}$ and the $Ran(f^{-1}(x)) := \{y \in \mathbb{R} | x \leq 4\}$ and

$Dom(f^{-1}(x)) := \{x \in \mathbb{R} | x \geq 0\}$. Graphs of $f(x) = \sqrt{4-x}$ and $y = 4-x^2$ are reflections of each other in $y = x$

Example:

Find the inverse function of $f(x) = x^3 + 1$ and sketch the graphs of $y = f(x)$ and $y = f^{-1}(x)$.